

Executive Summary

Socio-Economic Workforce Analysis

Prepared for the Chief Human Resources Officer (CHRO)

Overview

This analysis evaluates key drivers of employee compensation, pay equity across demographic groups, and structural differences between departments. The findings are based on a comprehensive review of the organization's socio-economic dataset, including salary, experience, education, performance, demographic attributes, and work arrangements.

The goal is to provide clear, actionable insights to support equitable compensation strategies, workforce planning, and evidence-based HR decision-making.

Key Findings

1. Pay Equity Across Gender and Demographic Groups

- **Gender and Department are not statistically associated** when evaluated using Chi-Square tests. This indicates that representation across departments is not driven by gender-based sorting.
- **Mean salary differs significantly between Male and Female employees**, based on the independent t-test. This suggests a potential pay equity concern that warrants deeper review.
- **Job satisfaction does not differ significantly** between union and non-union employees, indicating consistent employee experience across these groups.
- **Salary distributions do not differ significantly across marital status groups**, suggesting no compensation bias linked to marital status.

Implication: While demographic representation appears equitable, compensation differences between genders require further investigation to determine whether they stem from role allocation, performance, tenure, or structural bias.

2. Departmental Compensation Differences

- Salary levels differ significantly across the five departments, as confirmed by one-way ANOVA.
- These differences likely reflect variations in job families, skill requirements, and market rates.
- Remote work adoption varies substantially by department, indicating operational differences in job design and flexibility.

Implication: Department-level salary differences are expected but should be monitored to ensure they align with market benchmarks and internal equity guidelines.

3. Employee Compensation Drivers

Regression analysis reveals the strongest predictors of salary:

Years of Experience

- A clear, positive relationship exists between experience and salary.
- Each additional year of experience increases predicted salary by approximately **\$870**, reinforcing the value of tenure and accumulated expertise.

Education Level

- Higher education levels are associated with higher compensation, reflecting skill depth and qualification requirements.

Performance Score

- Performance is a meaningful driver of salary, supporting a merit-based compensation structure.

Gender and Department Effects

- After controlling for experience, education, and performance, **Gender and Department still show measurable effects.**
- This suggests structural influences beyond individual qualifications and performance.

Model Strength

- The multiple regression model explains **approximately 51.5% of salary variation**, indicating strong explanatory power.
- Log-transformation improved model stability and reduced heteroscedasticity.

Implication: Compensation is driven by a combination of human capital factors (experience, education, performance) and organizational structure (department, role family). Gender-based differences persist even after adjustment, highlighting an area for policy review.

Recommendations

1. Conduct a Formal Pay Equity Audit

- Perform a deeper regression-based pay equity analysis controlling job role, tenure, performance, and department.
- Identify unexplained pay gaps and develop corrective action plans.

2. Standardize Compensation Bands Across Departments

- Ensure departmental salary differences reflect market benchmarks rather than legacy structures.
- Introduce periodic compensation calibration sessions.

3. Strengthen Performance-Based Compensation Policies

- Reinforce transparent performance evaluation criteria.
- Ensure performance scores are consistently applied across departments.

4. Enhance Workforce Planning Using Tenure and Experience Data

- Use tenure insights to forecast retirement risk, succession needs, and leadership pipeline strength.

5. Monitor Remote Work Equity

- Ensure remote work access does not inadvertently influence compensation or promotion opportunities.
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Conclusion

The organization demonstrates strong alignment between compensation and key human-capital drivers such as experience, education, and performance. However, gender-based salary differences and departmental disparities warrant further review to ensure full compliance with equitable pay principles.

By implementing the recommended actions, the CHRO can strengthen fairness, transparency, and strategic workforce planning across the organization.
